Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · Income & Employment Verification · RESTful APIs · Production Reliability · AWS · CI/CD · PostgreSQL · Docker · Mortgage Data Pipelines

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PROFESSIONAL SUMMARY

Senior Python Engineer with 8 years owning production income verification and compliance pipelines - structurally identical to what Truework builds for mortgage lenders. At Entrata, was the long-term owner of HUD income-limit import: federal source data → 180,000 domain records powering eligibility, rent, and HAP calculations for 100K+ housing units across 7 years, zero compliance violations. Income eligibility data at this scale and regulatory stakes is the same problem as employment income verification for mortgage origination - wrong output means wrong decision for a real person. Delivered this at 99.5% uptime via RabbitMQ async processing, FastAPI REST APIs, PostgreSQL, and AWS. Built and shipped two AI systems from experimentation through production hardening: Leasel (8-agent LangGraph platform, 93.6% precision) and AgentProbe (RAG evaluation, CI-integrated regression gate). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Engineering & Backend: Python (8yr), SQL, FastAPI, Flask, PHP, JavaScript, TypeScript, Bash, Docker, Microservices, REST APIs, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue), RabbitMQ, Redis, PostgreSQL, MySQL, Oracle, Linux, Git

Data & Compliance Pipelines: Income Verification Pipelines, ETL, Data Validation & Reconciliation, Feature Engineering, Schema Design, Execution Plan Analysis, Query Optimization, Partitioning, Indexing, Stored Procedures, PHI Compliance (Libsodium), PySpark, NumPy, Pandas, Tableau

AI & Observability: LangGraph, LangChain, RAG Pipelines, LLM Orchestration, ChromaDB, Model Evaluation, HITL, Prompt Engineering, Observability, Production Monitoring, Agile/Scrum, BERT, spaCy, scikit-learn, PyTorch

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built agentic Python backend (LangGraph) for mission planning - rapid MVP experimentation to validate compliance logic, then iterated to production-hardened system with validated, cited outputs; CI/CD-integrated automated testing enforcing correctness at every release.
- Implemented RAG retrieval over distributed data sources with customer-centric validation guardrails; observability on retrieval quality and response groundedness across edge cases.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built Python and R ML pipelines on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through feature engineering, model evaluation, and statistical validation across 190+ countries.
- Mentored 600+ students in Python, SQL, and data science; designed monitoring dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned income verification pipeline end-to-end for 7 years** - annual HUD income-limit import: federal source data → 180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units; baseline validation, reconciliation, and rollback preparation at every release; zero compliance violations. The same income data integrity problem Truework solves for mortgage lenders.
- **Built and operated production REST APIs and async processing at scale** - FastAPI backends, RabbitMQ async processing (retry, idempotency, dead-letter) for gross-rent updates across thousands of tenant records; Redis caching; 99.5% uptime SLA; monitoring and triage tooling reducing resolution time 40%.
- **Python backend and database engineering** - 200+ SQL/PostgreSQL/Oracle query optimizations (30% performance gain) through execution plan analysis, indexing, partitioning, and stored procedure refactoring; developed secure, performant data pipelines with 100% PHI compliance via Libsodium encryption (zero security incidents); AWS S3, EC2, RDS at production scale.
- **Full ownership of 20+ scalable, reliable production modules** - Docker microservices redesign (50% scalability improvement, 30% deployment reduction); Python/Bash automation eliminating 80% of manual operational workload; 300+ code reviews with thorough documentation standards; mentored 5+ engineers on backend and data engineering best practices.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built income and housing compliance backend modules (PHP, Python, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records with validation and reconciliation; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

Leasel: Agentic AI Support Investigation Platform Agentic · RAG

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production agentic AI system for compliance investigation: 8-agent LangGraph stateful workflow, multi-source RAG with citation grounding, HITL escalation when model confidence is below threshold, and deterministic logic controlling every regulated output. 93.6% precision, 87.9% F1 across 1,240 residential compliance test cases; CI quality gate with telemetry-driven scoring.

AgentProbe: Production RAG Evaluation & Observability Platform Evaluation · CI/CD

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Python automated evaluation platform: 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate blocking failures before merge, LLM-as-judge scoring. Zero regressions shipped across 1,240+ test cases.

AI on FHIR: NLP-Driven Medical Records Query System Python · NLP

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Python + React platform translating natural-language queries into FHIR API calls via BERT embeddings and LLMs - reduced retrieval time ~60%; HIPAA-compliant RBAC with full data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017